

HKCEE 1994 Mathematics II

Label 1 If $f(x) = x^2 + 2x$, then $f(x-1) =$

- A. x^2
- B. $x^2 - 1$
- C. $x^2 + 2x - 1$
- D. $x^2 + 2x - 3$
- E. $x^2 + 4x - 1$

Label 2 If $y = \frac{2x-1}{x+2}$, then $x =$

- A. $\frac{1+3y}{2}$
- B. $\frac{1+2y}{2+y}$
- C. $\frac{1+2y}{2-y}$
- D. $\frac{1-2y}{2+y}$
- E. $\frac{1-2y}{2-y}$

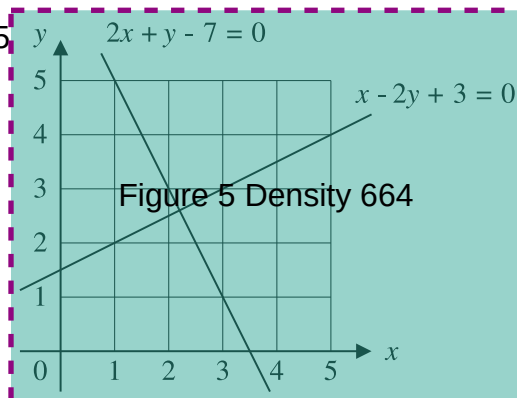
Label 3 The L.C.M. of $(x-1)^2$, x^3-1 and x^3-1 is

- A. $x-1$
- B. $(x-1)^4(x+1)(x^2+x+1)$
- C. $(x-1)^2(x+1)(x^2+x+1)$
- D. $(x-1)^2(x+1)(x^2-x+1)$
- E. $(x-1)(x+1)(x^2+x+1)$

Label 4 If $a = \sqrt{3} + \sqrt{2}$, then $a - \frac{1}{a} =$

- A. 0
- B. $2\sqrt{2}$
- C. $2\sqrt{3}$
- D. $\sqrt{3} - \sqrt{2}$
- E. $\frac{2\sqrt{3}}{3} + \frac{\sqrt{2}}{2}$

Label 5



In the figure, (x, y) is a point in the shaded region (including the boundary) and x, y are integers. Find the greatest value of $3x + y$.

- A. 7
- B. 8
- C. 9.2
- D. 10
- E. 10.5

Label 6 If $x(x+1) < 5(x+1)$, then

- A. $x \leq 5$
- B. $x < -5$ or $x > 1$
- C. $x < -5$ or $x > 1$
- D. $-5 < x < 1$
- E. $-1 < x < 5$

Label 7 Which of the following is/are an identity/identities?

- I. $(x+2)(x-2) = x^2 - 4$
- II. $(x+2)(x-2) = 0$
- III. $(x+2)^3 = x^3 + 8$

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

Label 8: If $\alpha \neq \beta$ and $\begin{cases} 3\alpha^2 - h\alpha - b = 0 \\ 3\beta^2 - h\beta - b = 0 \end{cases}$ then $\alpha + \beta =$

- A. $-\frac{b}{3}$
- B. $\frac{b}{3}$
- C. h
- D. h
- E. h

Label 9: Mr. Chan bought a car for \$ 143 900. If the value of the car goes down by 10% each year, find its value at the end of the third year. (Give your answer correct to the nearest hundred dollars.)

- A. \$ 94 400
- B. \$ 100 700
- C. \$ 104 900
- D. \$ 115 100
- E. \$ 116 600

Label 10: A wholesaler sells an article to a retailer at a profit of 20 %. The retailer sells it to a customer for \$ 3 600 at a profit of \$ 720. Find the original cost of the article to the wholesaler.

- A. \$ 2 304
- B. \$ 2 400
- C. \$ 2 880
- D. \$ 3 000
- E. \$ 3 456

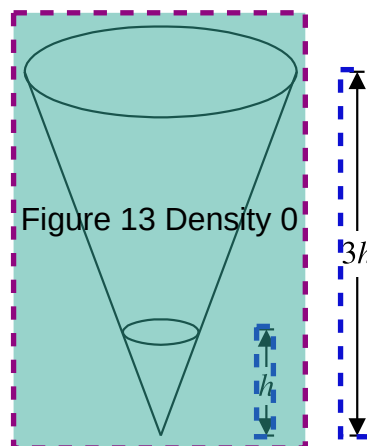
Label 11: The bearing of A from B is 075° . What is the bearing of B from A?

- A. 015°
- B. 075°
- C. 105°
- D. 195°
- E. 255°

Label 12: If the sum to infinity of a G.P. is $\frac{81}{4}$ and its second term is -9 , the common ratio is

- A. $\frac{1}{3}$
- B. $\frac{1}{3}$
- C. $\frac{4}{3}$
- D. $\frac{4}{3}$
- E. $\frac{4}{9}$

Label 13



In the figure, the paper cup in the form of a circular cone contains 10 ml of water. How many ml of water must be added to fill up the paper cup?

- A. 20
- B. 80
- C. 90
- D. 260
- E. 270

Label 14

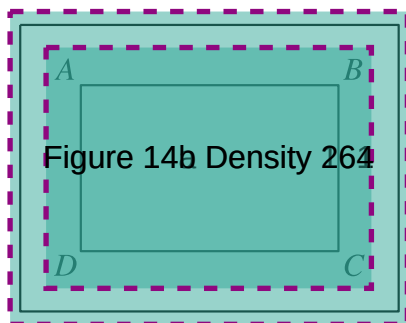


Figure 14b Density 264

In the figure, $ABCD$ is a rectangular field of length p metres and width q metres. The path around the field is of width 2 metres. Find the area of the path.

- A. $(4p + 4q) \text{ m}^2$
- B. $(2p + 2q + 4) \text{ m}^2$
- C. $(2p + 2q + 16) \text{ m}^2$
- D. $(4p + 4q + 16) \text{ m}^2$
- E. $(pq + 4p + 4q + 16) \text{ m}^2$

Label 15

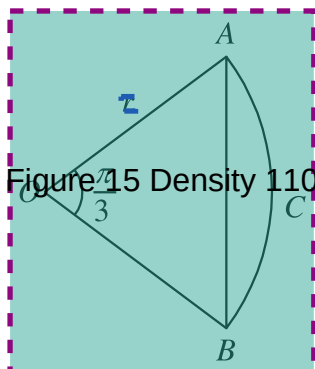


Figure 15 Density 110

In the figure, $OACB$ is a sector of radius r . If $\angle AOB = \frac{\pi}{3}$, find the area of the shaded part.

- A. $\left(\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right) r^2$
- B. $\left(\frac{\pi}{6} - \frac{1}{4}\right) r^2$
- C. $\left(\frac{\pi}{3} - \frac{\sqrt{3}}{2}\right) r^2$
- D. $\left(\frac{\pi}{3} - \frac{1}{2}\right) r^2$

E. $\frac{\pi}{3} r - \frac{\sqrt{3}}{4} r^2$

Label 16 $\frac{\cos \theta}{\sin \theta + 1} - \frac{\cos \theta}{\sin \theta - 1} =$

- A. $\frac{2}{\cos \theta}$
- B. $-\frac{2}{\cos \theta}$
- C. 0
- D. $2 \tan \theta$
- E. $-2 \tan \theta$

Label 17 Which of the following figures shows the graph of $y = 1 + \sin x$

A.

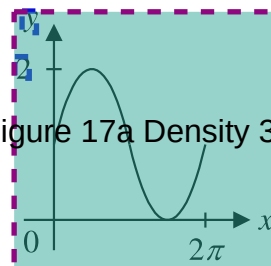


Figure 17a Density 361

B.

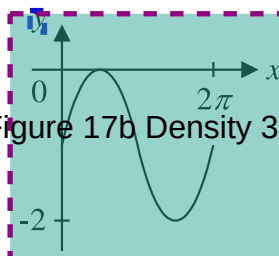


Figure 17b Density 363

C.

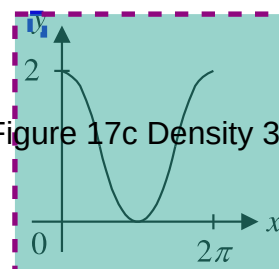


Figure 17c Density 351

D.

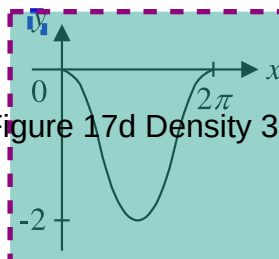


Figure 17d Density 359

E!

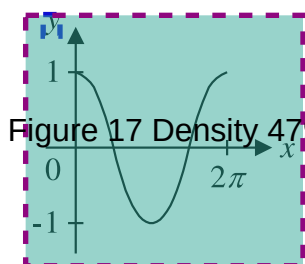


Figure 17 Density 479

Label 18

$$\frac{\sin(180^\circ + \theta)}{\cos(90^\circ - \theta)} =$$

- A. $\tan \theta$
- B. $-\tan \theta$
- C. $\frac{1}{\tan \theta}$
- D. $\frac{1}{\tan \theta}$
- E. $-\frac{1}{\tan \theta}$

Label 19

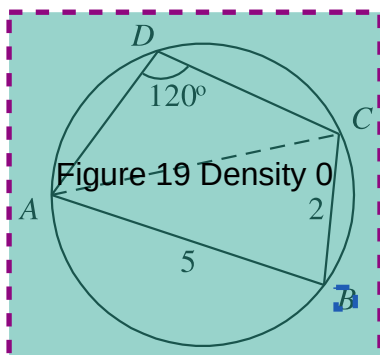


Figure 19 Density 0

In the figure, $ABCD$ is a cyclic quadrilateral with $AB = 5$, $BC = 2$ and $\angle ADC = 120^\circ$. Find AC .

- A. $\sqrt{19}$
- B. $\sqrt{21}$
- C. $\sqrt{6}$
- D. $\sqrt{34}$
- E. $\sqrt{39}$

Label 20

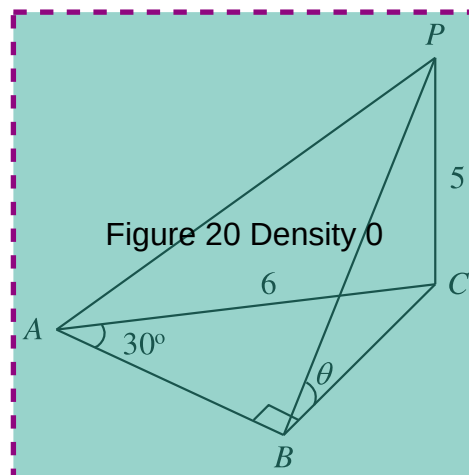


Figure 20 Density 0

In the figure, PC is a vertical pole standing on the horizontal plane ABC . If $\angle ABC = 90^\circ$, $\angle BAC = 30^\circ$, $AC = 6$ and $PC = 5$, find $\tan \theta$.

- A. $\frac{3}{5}$
- B. $\frac{5}{6}$
- C. $\frac{5}{3}$
- D. $\frac{3\sqrt{3}}{5}$
- E. $\frac{5\sqrt{3}}{9}$

Label 21

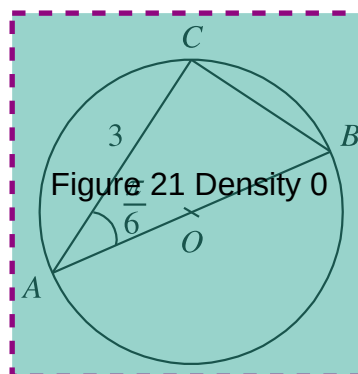


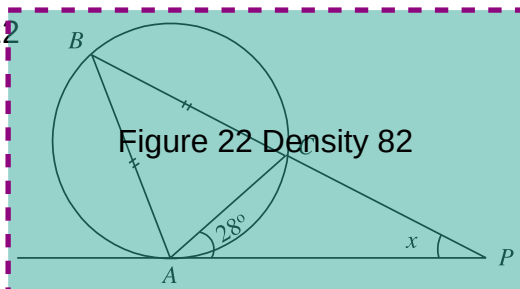
Figure 21 Density 0

In the figure, O is the centre of the circle. If $AC = 3$ and $\angle BAC = \frac{\pi}{6}$, find the diameter AB .

- A. $\frac{3}{2}$

- B. 6
C. $\frac{3\sqrt{3}}{2}$
D. $2\sqrt{3}$
E. $3\sqrt{3}$

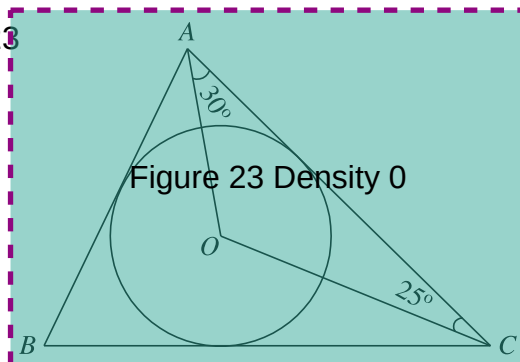
Label 22
22



In the figure, PA is tangent to the circle at A , $\angle CAP = 28^\circ$ and $BA = BC$. Find x .

- A. 28°
B. 48°
C. 56°
D. 62°
E. 76°

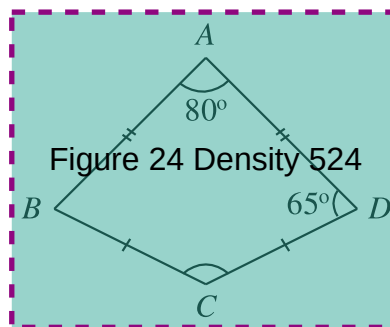
Label 23
23



In the figure, O is the centre of the inscribed circle of $\triangle ABC$. If $\angle OAC = 30^\circ$ and $\angle OCA = 25^\circ$, find $\angle ABC$.

- A. 50°
B. 55°
C. 60°
D. 62.5°
E. 70°

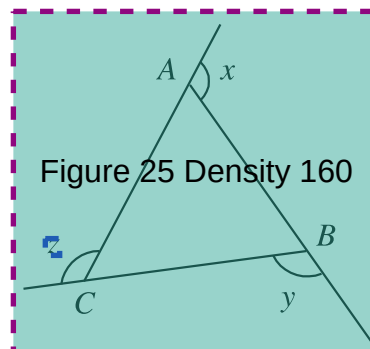
Label 24
24



In the figure, $AB = AD$ and $BC = CD$. If $\angle BAD = 80^\circ$ and $\angle ADC = 65^\circ$, then $\angle BCD =$

- A. 100°
B. 130°
C. 145°
D. 150°
E. 160°

Label 25
25



In the figure, x , y and z are the exterior angles of $\triangle ABC$. If $x : y : z = 4 : 5 : 6$, then $\angle BAC =$

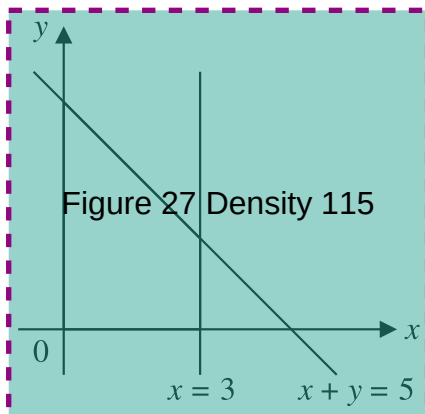
- A. 48°
B. 84°
C. 96°
D. 120°
E. 132°

Label 26
26

The points $A(4, -1)$, $B(-2, 3)$ and $C(x, 5)$ lie on a straight line. Find x .

- A. -5
B. -4
C. 0
D. 2
E. 5

Label 27
27



In the figure, the shaded part is bounded by the axes, the line $x = 3$ and $x + y = 5$. Find the area.

- A. 10.5
- B. 12
- C. 15
- D. 19.5
- E. 21

Label 28
28

AB is a diameter of the circle $x^2 + y^2 - 2x - 2y - 18 = 0$. If A is $(3, 5)$, then B is

- A. $(2, 3)$.
- B. $(1, -1)$.
- C. $(-1, -3)$
- D. $(-5, -7)$.
- E. $(-7, -9)$

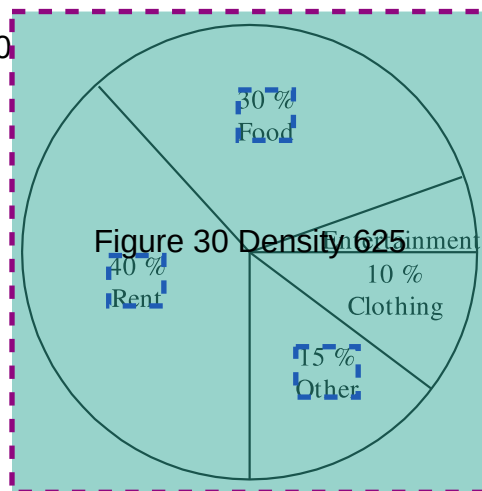
Label 29
29

The equations of two circles are $x^2 + y^2 - 4x - 6y = 0$,
 $x^2 + y^2 + 4x + 6y = 0$,
Which of the following is/are true?

- I. The two circles have the same centre.
- II. The two circles have equal radii.
- III. The two circles pass through the origin.

- A. I only
- B. II only
- C. III only
- D. I and III only
- E. II and III only

Label 30
30



In the figure, the pie chart shows the monthly expenditure of a family. If the family spends \$ 4 800 monthly on rent, what is the monthly expenditure on entertainment?

- A. \$240
- B. \$600
- C. \$720
- D. \$1 800
- E. \$12 000

Label 31
31

A box contains 5 eggs, 2 of which are rotten. If 2 eggs are chosen at random, find the probability that exactly one of them is rotten.

- A. $\frac{2}{5}$
- B. $\frac{3}{5}$
- C. $\frac{3}{10}$
- D. $\frac{6}{25}$
- E. $\frac{12}{25}$

Label 32
32

The mean, standard deviation and interquartile range of n numbers are m , s and q respectively. If 3 is added to each of the n numbers, what will be their new mean, standard deviation and interquartile range?

	Mean	Standard Deviation	Interquartile Range
A.	m	s	q
B.	m	$s \pm 3$	$q + 3$
C.	$m + 3$	s	q
D.	$m + 3$	s	$q + 3$
E.	$m + 3$	$s \pm 3$	$q + 3$

- C. $\frac{1}{2y - x}$
- D. $\frac{1}{2y + x}$
- E. $\frac{1}{4y - x}$

Label 33 $(3^x)^2 =$

- A. $3^{(x^2)}$
- B. 3^{x+2}
- C. 3^{2x}
- D. 6^x
- E. 9^{2x}

Label 34 If $\log 2 = a$ and $\log 9 = b$, then $\log 12 =$

- A. $2a + \frac{b}{3}$
- B. $2a + \frac{b}{2}$
- C. $\frac{2}{3}a + \frac{2}{3}b$
- D. $a^2 + b^{\frac{1}{2}}$
- E. $a^2 b^{\frac{1}{2}}$

Label 35 Factorize $a^2 - 2ab + b^2 - a + b$

- A. $(a - b)(a - b - 1)$
- B. $(a - b)(a - b + 1)$
- C. $(a - b)(a + b - 1)$
- D. $(a + b)(a - b - 1)$
- E. $(a - b - 1)^2$

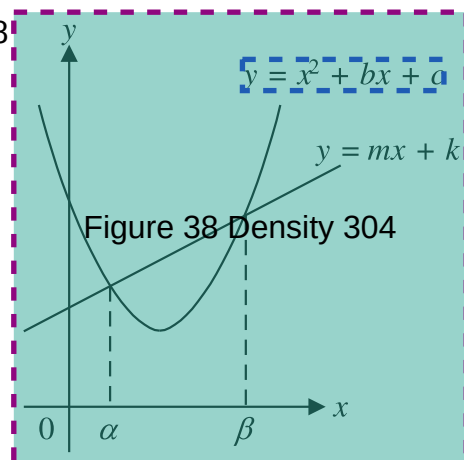
Label 36 $\frac{2}{x} - \frac{1}{y} =$
Figure 36 Density 935

- A. $2y - x$
- B. $2y + x$

Label 37 $P(x)$ is a polynomial. When $P(x)$ is divided by $(5x - 2)$, the remainder is R . If $P(x)$ is divided by $(2 - 5x)$, then the remainder is

- A. R .
- B. $-R$.
- C. $\frac{2}{5}R$.
- D. $\frac{2}{5}$.
- E. $-\frac{2}{5}$.

Label 38



In the figure, the line $y = mx + k$ cuts the curve $y = x^2 + bx + c$ at $x = \alpha$ and $x = \beta$. Find the value of $\alpha\beta$

- A. $-b$
- B. c
- C. $m - b$
- D. $k - c$
- E. $c - k$

Label 39
40 If $x = 3$, $y = 2$ satisfy the simultaneous equations $\begin{cases} ax + by = 2 \\ bx - ay = 3 \end{cases}$, find the values of a and b .

- A. $a = 0, b = 1$
- B. $a = 0, b = -1$
- C. $a = \frac{5}{6}, b = -\frac{1}{4}$
- D. $a = -\frac{1}{13}, b = \frac{37}{39}$
- E. $a = -\frac{12}{13}, b = \frac{5}{13}$

- B. $3 : 4 : 5$.
- C. $3 : 6 : 10$.
- D. $18 : 16 : 15$
- E. $40 : 45 : 48$

Label 43
43 Let x vary inversely as $\sqrt{y}$. If y is increased by 69%, then x will be

- A. increased by 23.1% (3 sig. fig.).
- B. increased by 30%.
- C. decreased by 23.1% (3 sig. fig.).
- D. decreased by 30%.
- E. decreased by 76.9% (3 sig. fig.).

Label 40
40 From the table, which of the following intervals must contain a root of $f(x) - x = 0$

x	$f(x)$
-2	1.2
-1	0.8
0	0.7
1	0.2
2	-0.1
3	0.8

- A. $-2 < x < -1$
- B. $-1 < x < 0$
- C. $0 < x < 1$
- D. $1 < x < 2$
- E. $2 < x < 3$

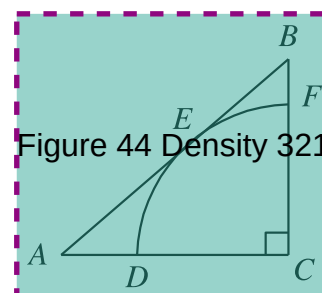
Label 41
41 If the product of the first n terms of the sequence $10, 10^2, 10^3, \dots, 10^n, \dots$ exceeds 10^{55} , find the minimum value of n .

- A. 9
- B. 10
- C. 11
- D. 12
- E. 56

Label 42
42 If $a : b = 2 : 3$, $a : c = 3 : 4$ and $a : d = 4 : 5$, then $b : c : d =$

- A. $2 : 3 : 4$

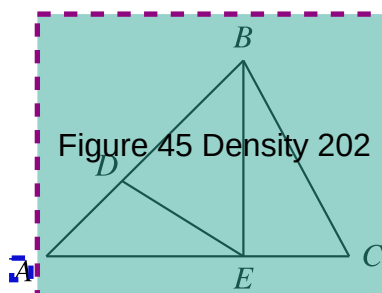
Label 44
44



In the figure, $CDEF$ is a sector of a circle which touched AB at E . If $AB = 25$ and $BC = 15$, find the radius of the sector.

- A. 9
- B. 10
- C. 11.25
- D. 12
- E. 12.5

Label 45
45



In the figure, $AD : DB = 1 : 2$, $AE : EC = 3 : 2$. Area of $\triangle BDE$: Area of $\triangle ABC =$

- A. $1 : 3$
- B. $2 : 5$

- C. 3 : 4
D. 4 : 25
E. 36 : 65

Label 46

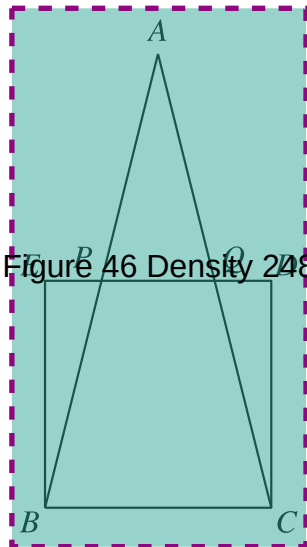


Figure 46 Density 248

In the figure, area of $\triangle ABC$: area of square $BCDE = 2 : 1$. Find $PQ : BC$.

- A. 1 : 2
B. 1 : 3
C. 1 : 4
D. 2 : 3
E. 3 : 4

Label 47 For $0^\circ \leq x \leq 360^\circ$, how many roots does the equation $\sin x(\cos x + 2) = 0$ have?

- A. 0
B. 1
C. 2
D. 3
E. 4

94 The largest value of $(3\cos 2\theta - 1)^2 + 1$ is

- A. 2
B. 5
C. 17
D. 26
E. 50

Label 49

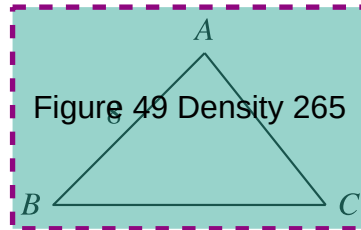


Figure 49 Density 265

In the figure, $\sin A : \sin B : \sin C = 4 : 5 : 6$. If $AB = 8$, find AC .

- A. $5\frac{1}{3}$
B. $6\frac{2}{3}$
C. $9\frac{3}{5}$
D. 10
E. 12

Label 50

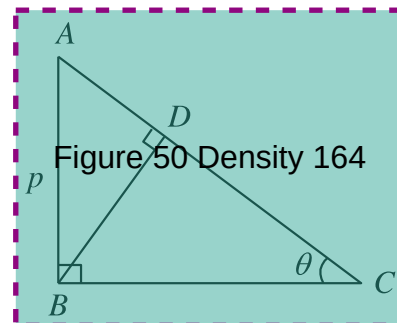


Figure 50 Density 164

In the figure, $AB = p$, $\angle ACB = \theta$. Find CD .

- A. $p \sin \theta$
B. $p \cos \theta$
C. $\frac{p \sin \theta}{\cos^2 \theta}$
D. $\frac{p \sin^2 \theta}{\cos \theta}$
E. $\frac{p \cos^2 \theta}{\sin \theta}$

Label 51

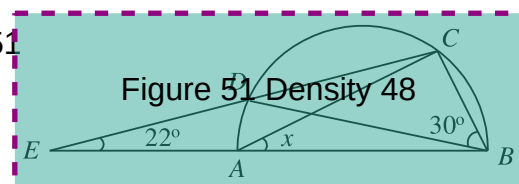


Figure 51 Density 48

In the figure, $ABCD$ is a semi-circle, CDE and BAE are straight lines. If $\angle CBD = 30^\circ$ and $\angle DEA = 22^\circ$, find x .

- A. 38°
- B. 41°
- C. 44°
- D. 52°
- E. 60°

- B. II only
- C. III only
- D. I and II only
- E. II and III only

Figure 54

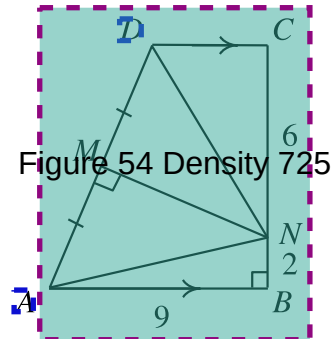


Figure 54 Density 725

In the figure, $ABCD$ is a trapezium with $AB \parallel DC$, $\angle ABC = 90^\circ$ and MN is the perpendicular bisector of AD . If $AB = 9$, $BN = 2$ and $NC = 6$, find CD .

- A. $4\frac{1}{2}$
- B. $6\frac{3}{4}$
- C. 7
- D. $\sqrt{41}$
- E. $\sqrt{113}$

Figure 52

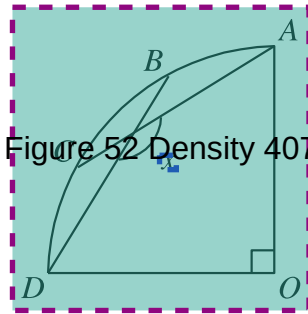


Figure 52 Density 407

In the figure, $OABCD$ is a sector of a circle. If arc $AB = \text{arc } BC = \text{arc } CD$, then $x =$

- A. 105°
- B. 120°
- C. 135°
- D. 144°
- E. 150°

Figure 53

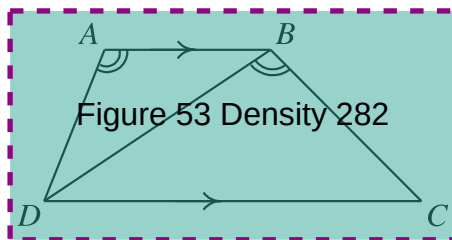


Figure 53 Density 282

In the figure, $AB \parallel DC$ and $\angle DAB = \angle DBC$. Which of the following is/are true?

- I. $\frac{AB}{BD} = \frac{BD}{DC}$
- II. $\frac{AB}{BD} = \frac{AD}{BC}$
- III. $\frac{AD}{BD} = \frac{BD}{CD}$

- A. I only